Multimodal Deep Fake Detection with Cross-Modal Attention and Explainable AI

A PROJECT REPORT

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ABSTRACT

Advancements in generative AI tools has increased the production and perception of deepfakes, threatening digital trust and information integrity. Dominant methods for detecting deepfakes have used individual modalities that address the problem from a simple binary classification point of view. This has often led to a lack of generalization among different data sets, which cannot capture the complex interactions between audio-visual signals that characterize modern deepfakes. Moreover, these methods are often not transparent since they are based on a black-box principle. In order to address these issues, we have proposed a multimodal deepfake detection framework that classifies deepfake media into different classes based on their manipulation type between the audio and visual signals. Our method involves extracting audio and video frames from the marked altered intervals of the media signals. This has reduced the effects of signal dilution and increased the focus on the altered parts. In order to increase the transparency of the proposed method, we have integrated a Large Language Model that provides clear forensic analysis describing the anomalies that characterize each classification decision. This has provided a generalizable paradigm for digital multimedia forensics.

Keywords: Deepfake Detection, Multimodal Fusion, Cross-Modal Attention, Explainable AI, Forensics, Audio-Visual Analysis

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LIST OF ACRONYMS AND ABBREVIATIONS

Abbreviation	Description
ResNet	Residual Network
AUC	Area Under Curve
ROC	Receiver Operating Characteristic
CNN	Convolutional Neural Network
GAN	Generative Adversarial Network
XAI	Explainable Artificial Intelligence

CHAPTER 1

INTRODUCTION

1.1 PROBLEM STATEMENT

The rapid evolution of AI technology has led to the emergence of numerous deepfakes that contain highly sophisticated video and audio manipulations that are hard to detect. Most deepfake detection algorithms have relied exclusively on single-modal methods of analysis, examining only the audio or video components of the media file. Although such approaches can produce accurate results in specific cases, they tend to underperform when analyzing different datasets containing various types of manipulations.

Another major limitation associated with most existing systems of detecting deepfakes is their use of black-box methods that allow the prediction to be made without explaining why the result was obtained. As a consequence, such approaches cannot be reliably used in many important applications related to digital forensics and media security.

In particular, contemporary deepfakes feature multimodal manipulation, which means that both video and audio streams are modified in ways that do not fit into a temporal pattern. Traditional detection systems tend to be ineffective in detecting inconsistencies between the modalities of manipulated media files.

1.2 AIM OF THE PROJECT

- Designing a multimodal technique to identify deepfakes by making use of visuals and sounds to improve detection.
- Synthesizing an algorithm that is able to categorize deepfakes based on their manipulation techniques.
- Gaining robustness by training the algorithm on multiple sets of data that have distinct characteristics.
- Applying explainable AI in order to make the reasoning process understandable to humans.

1.3 PROJECT DOMAIN

This topic of study falls under the disciplines of Artificial Intelligence, Deep Learning, and Multimedia Forensics. More specifically, the topic revolves around the application of multimodal learning techniques in the examination of visual and auditory content in order to identify the existence of any artificial media. The topic also takes into consideration that of Explainable AI (XAI).

1.4 SCOPE OF THE PROJECT

The scope of this project covers the creation and deployment of a multimodal deepfake detection system that analyzes both audio and visual aspects of video content. Further, this will incorporate a time-stamp-based preprocessing phase that zeroes in on modified portions, as well as cross-modality attention capabilities for identifying contradictions across the audio and visual channels. Moreover, this will come equipped with an explainability component which creates explanatory forensic reports that can help with the practical applications of this system.

1.5 METHODOLOGY

This system is designed following a multimodal deep learning method for deepfake detection. Initially, video inputs undergo preprocessing to extract frames for faces and audio components. The visual modality will be analyzed by a convolutional neural network, while the audio modality will be analyzed by mel-spectrograms. Attention is then applied on both modalities, where the system can learn interactions between audio and visual.

The output from the attention mechanism is fed into a classifier that identifies whether the input video is real, video fake, audio fake, or fully fake. To increase the level of model interpretability, explanation techniques are applied, including spatial attention, temporal analysis, and audio saliency. This output is then further processed by a language model to produce a human-understandable forensic report of the identified anomalies.

1.6 ORGANIZATION OF THE REPORT

The contents of chapter 1 will be an introduction of the research, including the statement of the problem, objective, scope, and methodology of the research. Chapter 2 will have a literature review on the subject matter, where there will be a review of existing solutions to the problems of detecting deepfakes and their limitations. Chapter 3 will focus on the current system and proposed solution, including the feasibility analysis and specification of the proposed solution.

Chapter 4 will discuss the structure and architecture of the system under development, including discussions about the processes involved in pre-processing, features extraction, multimodal fusion, and explainability. Chapter 5 will cover the testing of the system, where different ways in which tests will be carried out will be highlighted. Chapter 6 will present the results obtained from the tests of the proposed solution, including analyses of the accuracy, AUC, and error rates.

CHAPTER 2
LITERATURE REVIEW

The Raza et al. study proposed a hybrid model called Deep Fake Prediction (DFP) that combines transfer learning with custom CNN layers for the automated extraction of key facial features for classification [1]. The proposed method recorded a high accuracy rate of 94% and 95% precision on facial data sets compared to other traditional models like Xception, NAS-Net, and MobileNet. This study proves that a combination of transfer learning improves classification results while controlling overfitting for forensic investigators. In another study, Mubarak et al. undertook a wide-ranging survey on different detection mechanisms for visual, audio, and text media [2]. The study highlighted the underestimated risks that LLM-generated deep fakes pose for forensic investigators. After examining 255 papers on deep learning-based deep fake detection methods, the study concluded that although high accuracy rates are achievable for a limited data set, cross-dataset generalization is a major challenge for forensic investigators. This study proves that there is a pressing need for flexible systems that can detect subtle manipulations in different digital media.

Abdulhamied et al. have attempted to leverage the power of audio-video multimodal fusion to enhance the reliability of forensic systems, especially under challenging cross-manipulation conditions where the performance of unimodal detectors is typically low [3]. By utilizing the CNN-LSTM audio features and the ResNet-50 visual features, the proposed model achieved a high AUC of 0.97 on the same manipulations and 0.72 on novel manipulations not seen during training. This demonstrates the power of multimodal fusion to leverage the cross-modal consistencies to achieve robustness over manipulations. Mathews et al. have proposed the DFIM-HQ dataset along with a MesoInception-Net architecture incorporating the power of Grad-CAM to provide explainability to the decision-making process of the deepfake detection model [4]. The proposed model achieved a high accuracy of 99.87%. This demonstrates the importance of Explainable AI (XAI) as a fundamental requirement to develop reliable forensic models that can provide valuable insights into the irregular forgery features embedded in heterogeneous digital media.

The study by Rafique et al. examined the possibility of combining ELA with ResNet18 and SqueezeNet for detecting discrepancies between pixels in face images [5]. The study was able to attain a level of accuracy close to 89.5% when checking static face images, but it also recognizes one of its major drawbacks: ELA-based spatial detection is not effective when it comes to checking videos, owing to their dynamic nature. The study concludes that this detection method is better suited for image forensics but not for the dynamic world of video-based deepfakes. Sunkari et al., on the other hand, developed "DeepExplain," a hybrid CNN-LSTM model that focuses on increasing transparency in video forensics [6]. This model is able to identify patterns of inconsistency between frames through spatial and temporal analysis, obtaining an accuracy of up to 97%. The team also utilized Grad-CAM for increased understanding of their model by humans, emphasizing that it is time to break away from 'black box' approaches in cybersecurity and forensics. Researchers Rafique, et al., attempted a combination of Error Level Analysis (ELA) with light transfer learning, employing ShuffleNet for image forgery detection with lower computational cost [5]. The technique detects minor inconsistencies caused by compression, followed by the extraction of deep discriminative features, achieving 88.2% accuracy with a KNN classifier. This proves the effectiveness of the ELA preprocessing technique for image tampering detection, with the transfer learning aspect contributing to the efficiency of the technique for handling large amounts of data in real-time.

Malik, along with the research team, analyzed the state-of-the-art deep learning techniques for creating deepfakes, as well as detecting them, focusing on the detection of facial alterations with the aid of spatial artifacts and temporal irregularities [7]. The research concluded that, although deep learning architectures can achieve near-perfect results, their effectiveness drops to zero when tested with adversarial datasets. The conclusion drawn from the research is that multimedia forensic techniques must shift their focus from the current domain adaptation techniques to accommodate the rapid developments in the field of hyper-realistic image/video generation with the aid of deep learning techniques. In a survey of 132 papers, Kaur et al. highlighted the challenges in video forensics, particularly in keeping frames consistent with each other, as well as audio with video [8]. While deep models like XceptionNet can attain high AUC-ROC results, they are still challenged when presented with black box, never-before-seen datasets in video forensics. The takeaway here is obvious: we need more efficient learning as well as better explainability if we are to go beyond what current data-based models can attain.

In a different paper, Hamza et al. investigated synthetic speech detection by feature extraction via Mel Frequency Cepstral Coefficients (MFCC), then inputting these features into machine learning algorithms like SVM and RF [9]. While they were able to attain a high accuracy of 98.83% via SVM on the Fake or Real dataset, their paper highlights how important these features are in distinguishing between human and artificially generated speech. Bekheet et al. proposed a multimodal framework called MAVDD that combines the analysis of different facial regions in images along with other robust audio cues like Mel-spectrograms and MFCCs [10]. The combination of all these features has yielded positive outcomes when distinguishing between authentic media and deepfake videos where both audio and visual segments were manipulated. This was made clear by the observation that the proposed framework outperformed all other methods tested on a single modality basis, thereby confirming that using Xception features together with MFCCs was the most effective way to represent media forensics.

Summing up all the above considerations, it should be stated that the discussed studies show significant progress in designing deepfake detection systems using deep learning and multimodal analysis. The effectiveness of convolutional neural networks, transformer models, and cross-modal methods in detecting manipulations created by advanced technologies is proven. Additionally, utilizing explainable AI provides greater transparency of the designed solutions. Yet, the described drawbacks of existing systems demonstrate that there remains a need for improvement of multimodal approaches providing high accuracy and transparency simultaneously.

CHAPTER 3

PROJECT DESCRIPTION

3.1 EXISTING SYSTEM

The detection of deepfakes becomes increasingly difficult as technologies used to create deepfakes evolve. In most cases, the deepfake video detection system utilizes specialized neural network architectures capable of recognizing manipulations and errors in video frames. However, current deepfake creators have mastered the technology to the point where they rarely leave any traces of manipulation in the video frames. In regard to the audio, the detection system typically focuses on the frequency and other features of sound waves to identify manipulations, yet they prove ineffective at detecting minor distortions of the speaker's voice.

The most significant limitation of existing solutions lies in their inability to operate on multiple data sets. A neural network trained on a particular data set performs perfectly while working on it but fails miserably with another data set. Furthermore, many detectors based on deep learning operate as black boxes meaning that their predictions can hardly be understood by anyone. Therefore, such detectors lack reliability as they provide no explanations of their decisions. In addition, it prevents users from using the system in criminal investigations since it may help to find out more information about what happened than what deepfake detection alone does. Therefore, this lack of transparency and interpretability is a crucial flaw of existing deepfake detectors. Therefore, modern deepfake detectors are constrained by poor support for multimodality, non- transparency, and low performance in practice

.

3.2 PROPOSED SYSTEM

A multimodal deepfake detection technique will be implemented in the new architecture, which utilizes both the visual and audio modalities to achieve accurate detections. In addition, the model uses a cross-modal learning process, whereby video frames are analyzed simultaneously alongside their respective audio streams.

The process starts by analyzing the input video to extract the facial frames and the corresponding audio streams. In particular, timestamp-aware sampling will be used to filter out non-manipulated frames and thus concentrate on the manipulation areas.

- Visual Features Extraction Process: Frames are analyzed by a CNN-based model (ResNet-encoder) to get spatial features of the manipulated video faces. Such a process enables the model to identify any visual anomalies in the manipulated video.

- Audio Features Extraction: The input audio stream is first transformed to mel spectrogram, then processed with the CNN-based architecture to get the audio features. Such a process helps in identifying the presence of synthetic voices.

- Cross-Modal Feature Fusion: Both the visual and audio features are fused via cross-modal learning techniques to identify any inconsistencies across the modalities. For instance, an inconsistency may include the mismatch between lip sync and audio content.

- Classification: The combined features are then sent to the classification stage where they are classified into one of the four categories – real, video manipulation, audio manipulation, or full manipulation.

- Explainability Module: To increase transparency, explainable AI is used in this process via methods like Grad-CAM for spatial interpretation, temporal attention to understand the significance of frames, and audio saliency detection. The results are then fed into a language model that converts them into forensic reports in human-readable language.

3.2.1 ADVANTAGES

The novel deepfake detection framework has several key strengths compared to existing frameworks. The first strength is its ability to analyze multimodal inputs – both visual and auditory data simultaneously. This allows detecting highly sophisticated deepfakes that cannot be detected by analyzing only one modality. Thus, the proposed framework outperforms other methods in detecting multimodal deepfakes.

Another strength of the novel method is its effectiveness in dealing with a large variety of datasets. When the framework is trained using a combination of multiple unique datasets, it gains the capability to see the big picture, thus generalizing knowledge that is acquired on multiple distinct datasets rather than learning patterns that are unique to one particular dataset. The framework gains the capability to generalize knowledge, and it can be applied to a wide range of problems.

Explaining the process through explainable AI tools increases transparency. The algorithm is now clear and predictable. It can use visuals, audios, and timings to justify predictions and provide further explanation on how it reached specific results.

3.3 FEASIBILITY STUDY

The procedure entails conducting a feasibility study, which is essential in ensuring that the project will work. It involves assessing its feasibility in terms of economics, technology, and social aspect.

3.3.1 ECONOMIC STUDY

The suggested approach will be economically viable because it uses freely accessible data sets and open-source technologies in its implementation. Deep learning frameworks, such as PyTorch and TensorFlow, will allow us to avoid paying for proprietary software. Of course, training a neural network requires substantial computing resources, including a graphic processor, but this task does not require considerable expenses.

Moreover, it should be noted that after the training process, the model can produce predictions without the help of any high-end hardware, making it very cost-effective. There will be no expenses related to the acquisition of licenses, while open-source software will ensure affordability of the suggested system for research organizations and academic projects.

3.3.2 TECHNICAL FEASIBILITY

From a technical point of view, such a model can definitely be implemented successfully. Indeed, there are numerous powerful tools currently used by the industry to create and implement machine learning applications. Specifically, state-of-the-art deep learning architecture combined with efficient

multimedia processing algorithms will allow extracting necessary information from audio and video content very efficiently. The ability to apply convolutional neural networks and transformer-based attention mechanisms will contribute to a very high learning efficiency and enable the system to focus on essential features. Overall, from a technological perspective, developing such a model appears fully realistic. The implementation of such a model requires proper data preparation. When dealing with face recognition and voice recognition models, you need to preprocess your datasets to ensure they are suitable for training. There are numerous valuable libraries available that will help you with data preparation. For instance, OpenCV can help you with face detection, and you can use Librosa library to prepare audio data. Moreover, the implementation of the model using several databases will enhance its ability to generalize and perform well. Moreover, the design of this model is modular; therefore, its advantages include scalability and testability among others.

3.3.3 SOCIAL FEASIBILITY

This solution is extremely relevant in the context of the current situation within our society because it focuses on solving the problem associated with the proliferation of deepfake content that is a growing issue in today's digital world. First of all, deepfake media can be used to deceive people, steal their identities, and violate their privacy. This creates a serious threat not only to individual users but to the entire community as well. Therefore, through the creation of a detection mechanism, the project will help build confidence in the use of media in a digital world and promote ethical use of artificial intelligence technology. In addition, due to the inclusion of an explainability module, the proposed system will comply with legislation, making its use in forensic analysis possible. An important feature of this solution is the use of open-source tools in order to benefit society in general.

3.4 SYSTEM SPECIFICATION

The multimodal deepfake detector that would be designed in this work relies on deep learning approaches, multimedia analysis techniques, and explainability approaches to construct an effective detection framework suitable for video and audio databases. In particular, the proposed solution involves a visual encoder designed for extraction of spatial information from visual data, an audio encoder, which works with audio data, and a cross-modal fusion mechanism used for fusion of information extracted from two modalities.

3.4.1 HARDWARE SPECIFICATION

The hardware requirements for the proposed system are primarily focused on supporting deep learning training and inference tasks.

- Processor: Multi-core CPU (Intel i5/i7 or equivalent)
- GPU: NVIDIA GPU (recommended for training, e.g., RTX series)
- RAM: Minimum 8 GB (16 GB recommended for smoother processing)
- Storage: Minimum 50 GB for datasets and model storage
- Additional: High-speed internet for dataset access and model training

These specifications ensure efficient handling of large-scale datasets, model training, and inference operations.

3.4.2 STANDARDS AND POILICES

To guarantee the reliability and ethical application of the proposed system, we will adhere to best practices in artificial intelligence and cybersecurity.

Security and Ethical Standards:

- Compliance with ethical principles in the use of deep learning algorithms.
- Conformity with privacy requirements during multimedia data processing.

Operational Policies:

- Data Handling Policy: Ensuring proper preprocessing and anonymization of datasets.
- Model Update Policy: Frequent retraining with new datasets to maintain detection accuracy.
- Evaluation Policy: Constant performance evaluation using metrics such as accuracy, AUC, and EER

Maintenance Policies:

- Regular updates to model architecture and preprocessing techniques
- Validation of system performance after each update
- Proper documentation of system components for reproducibility

CHAPTER 4

PROPOSED WORK

4.1 SYSTEM ARCHITECTURE

The proposed method involves building a multimodal deepfake detection system architecture involving processing of visual and audio data streams. As depicted in Figure 4.1, the process starts with feeding the system with a video file, after which the video undergoes a preprocessing phase where facial frames and respective audio signals are isolated from the video. In regard to the visual stream, spatial features are obtained by processing the video frames through a ResNet-based Convolutional Neural Network (CNN). On the other hand, the audio stream is converted into mel-spectrogram form, followed by extracting speech features using an audio encoder. The features from the two streams are then fed into a cross-modal transformer for analysis. Here, the attention mechanism helps establish relations between the visual and audio streams for detecting anomalies like lip sync. Moreover, the model calculates an audio-visual synchronization score. Lastly, the fused feature representation together with the synchronization score is classified into one of four labels, namely real, video manipulated, audio manipulated, or fully manipulated using the classification layer.

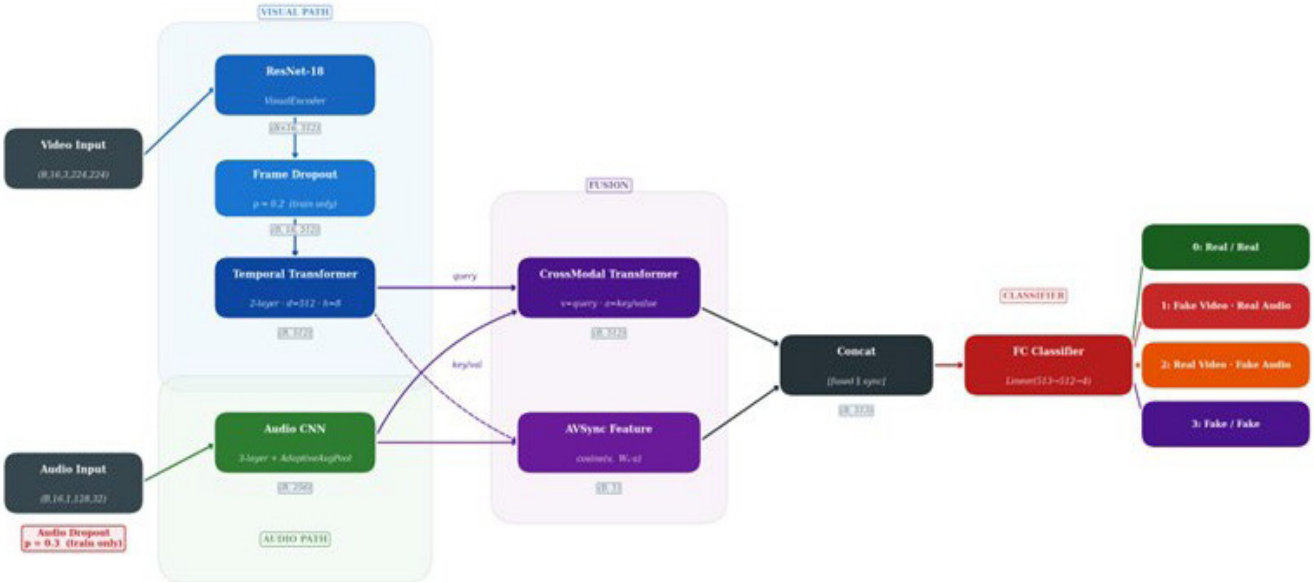

Figure 4.1: Architecture diagram

4.2 DESIGN PHASE

This stage is significant as it describes how the system for detecting multimodal deepfakes will process data and predict. The concept is turned into an implementation by breaking it down into modules that ensure efficient performance of the system. The design of the plan is crucial for its successful realization since it allows the system to perform successfully and with high efficiency in real life. In this case, the system will be able to operate in accordance with its goal.

Thus, it was decided to build the multimodal deepfake detector as a pipeline containing five modules: preprocessing, feature extraction, multimodal fusion, classification, and explainability modules. Every module is responsible for performing particular tasks and providing structured outputs to the following step.

To model the proposed system design, multiple diagrams like Data Flow Diagrams (DFD), Sequence Diagrams were selected. These types of diagrams allow depicting the movement of video and audio data in the system, inter-module interaction, generation of decision-making in detail.

In this case, it is necessary to use a Data Flow Diagram to present the process of converting raw videos into organized features and predictions while using a Sequence Diagram to display the sequence of processes from the receipt of input data up to obtaining an output result. By using the developed structure of the system, one may eliminate potential errors and increase efficiency of the system due to increased modularity.

4.2.1 DATA FLOW DIAGRAM

The Data Flow Diagram (DFD) shows the data flow in the multimodal deepfake detection system. The diagram provides information on how the input video data goes through different layers and finally produces results and explanations.

Firstly, the input video feeds into the preprocessing layer. In the preprocessing layer, faces are detected from the video, and audio signals are transformed to mel-spectrograms. The data output from the preprocessing layer is divided into two parallel streams for further processing:

- Visual Stream: Processes images using the convolutional neural network to obtain spatial features.
- Audio Stream: Processes mel-spectrograms with audio encoding to obtain audio features.

As depicted in figure 4.2, the obtained features are then sent to the multimodal fusion layer, where cross-modal attention mechanism helps to fuse features together into one feature vector.

The feature vector is passed to the classification layer, where the prediction is made on the class label. Besides, the explainability modules analyze the spatial-temporal and audio features to provide an explanation. The final outputs include both the classification result and a detailed forensic explanation, ensuring both accuracy and transparency.

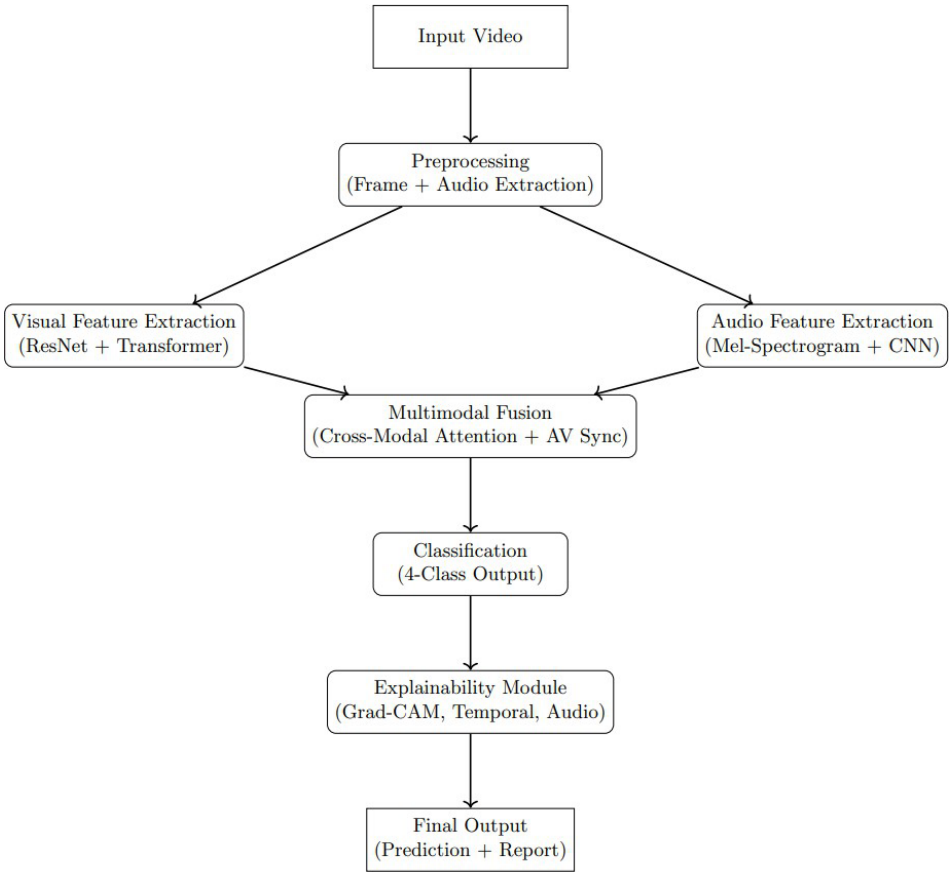

Figure 4.2: Dataflow diagram

4.3 MODULE DESCRIPTION

The suggested deepfake detection pipeline framework uses the concept of modules which play different roles in mapping the raw multimedia input to the final classified output with explainable justification. The proposed multimodal pipeline consists of the modules of data preprocessing, feature extraction, multimodal learning, and explainability in order to provide accurate and explainable results.

All modules are built based on the chosen methodology to help address the issues of generalization, inconsistency detection, and explainability.

4.3.1 MODULE 1: DATA PREPROCESSING AND DATASET HANDLING

The Data Preprocessing Module is an essential component that prepares the raw data from videos to be used for training and testing. The module works on three different datasets: FakeAVCeleb, LAV-DF, and FaceForensics++. There is a unified four-class classification label for all three datasets based on the presence of manipulation in video, audio, both, or none. This four-class label is more informative compared to the existing two-class labels.

Face Detection is achieved by RetinaFace. Only the largest face detected in each frame is used and followed consistently through all frames in the video. A total number of 16 frames are selected from each video, making sure that all videos have the same number of frames. However, for datasets like LAV-DF, timestamps are used to select only manipulated parts of the video to avoid diluting fake signals. The audio part of the video is extracted using FFmpeg, and then converted to mel-spectrograms using Librosa. The audio signal is split into multiple segments to capture temporal changes. If no audio is present, e.g., in FaceForensics++, dummy zero values are used. This module ensures that both visual and audio data are transformed into structured, standardized inputs suitable for deep learning models.

4.3.2 MODULE 2: VISUAL FEATURE EXTRACTION AND TEMPORAL MODELING

The Visual Feature Extraction Module analyzes the spatial and temporal information contained within the video frames. For the visual encoder, we use a pre-trained ResNet18 model, but remove its classification layer to generate high-level feature embeddings for each frame.

For a series of frames, the generated features are structured temporally and fed into the Temporal Encoder which is based on transformers. Through self-attention, long-term relations between frames can be learned, making it possible to detect temporal inconsistencies such as flickering effects and unrealistic motion sequences.

Figure 4.3: Extracted Visual Features

To enhance robustness during training, temporal dropout of some frame embeddings is employed. This way, it becomes impossible to exploit certain frames. The result of this module is a concise representation of spatial and temporal characteristics of the input video.

4.3.3 MODULE 3: AUDIO FEATURE EXTRACTION

The Audio Feature Extraction Module analyzes the audio part of the video to find any abnormalities in the speech pattern. The extracted audio signal is segmented into several parts and represented as mel-spectrogram features, including frequency and time information. Each mel-spectrogram is analyzed using a convolutional neural network to extract relevant audio features.

4.3.4 MODULE 4: MULTIMODAL FUSION AND CLASSIFICATION

These visual and audio representations can be combined using the Cross-Modal Attention mechanism available within the Multi-Modal Fusion layer. In contrast to approaches that utilize only concatenation operations, such mechanisms allow the recognition of correlation between two modalities since the visual representation can be used as the query while the audio representation acts as the key/value pair. This makes it possible for the model to detect differences including off-

sync movement of lips and mismatching voice data. The synchronization level of audio-visual content can be measured based on the cosine similarity between embeddings. After fusion, the resulting feature embedding and synchronization score is used as input for the classifier head comprising fully-connected layers. Finally, the network generates a four-class output representing the type of manipulation. This module acts as the primary decision-making module of the architecture.

MODULE 5: EXPLAINABILITY AND INFERENCE PIPELINE

The Explainability Module improves the system's interpretability by offering an in-depth explanation of the model's decision-making process. This module works on the inference stage and comprises three primary blocks. Spatial explainability is accomplished via Grad-CAM, which marks the parts of the video frame responsible for the model's prediction. The temporal attention block reveals the key frames in the sequence, suggesting locations of possible manipulation. The audio saliency block detects abnormal energy distribution in audio fragments, thus revealing synthetic speech. As depicted in Figure 4.3, All of these explainable outputs are concatenated and fed to a pre-trained large language model, producing a forensic report for a human user in understandable language.

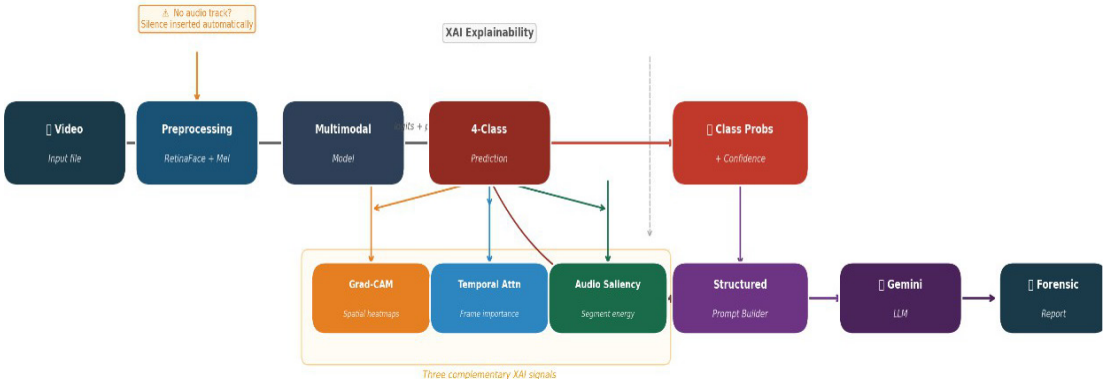

Figure 4.4: End-to-end inference and XAI pipeline

This module bridges the gap between deep learning predictions and human interpretation, making the system suitable for forensic and real-world applications.

CHAPTER 5

IMPLEMENTATION AND TESTING

5.1 INPUT AND OUTPUT

5.1.1 INPUT SAMPLE

- A sample video is selected from the dataset for evaluating the performance of the proposed multimodal deepfake detection system.
- The input video is processed through the preprocessing pipeline where facial frames and corresponding audio signals are extracted.
- The extracted frames are passed through the visual encoder, while the audio is converted into mel-spectrogram representations and processed through the audio encoder.

Figure 5.1 Input Frames Extracted from Video

5.1.2 MODEL OUTPUT AND EXPLAINABILITY

The processed input is passed through the multimodal model, which generates a prediction indicating whether the video is real or manipulated. Along with the prediction, the system provides explainability outputs including:

- Grad-CAM heatmaps highlighting important regions
- Temporal attention scores identifying important frames
- Audio saliency indicating suspicious audio segments

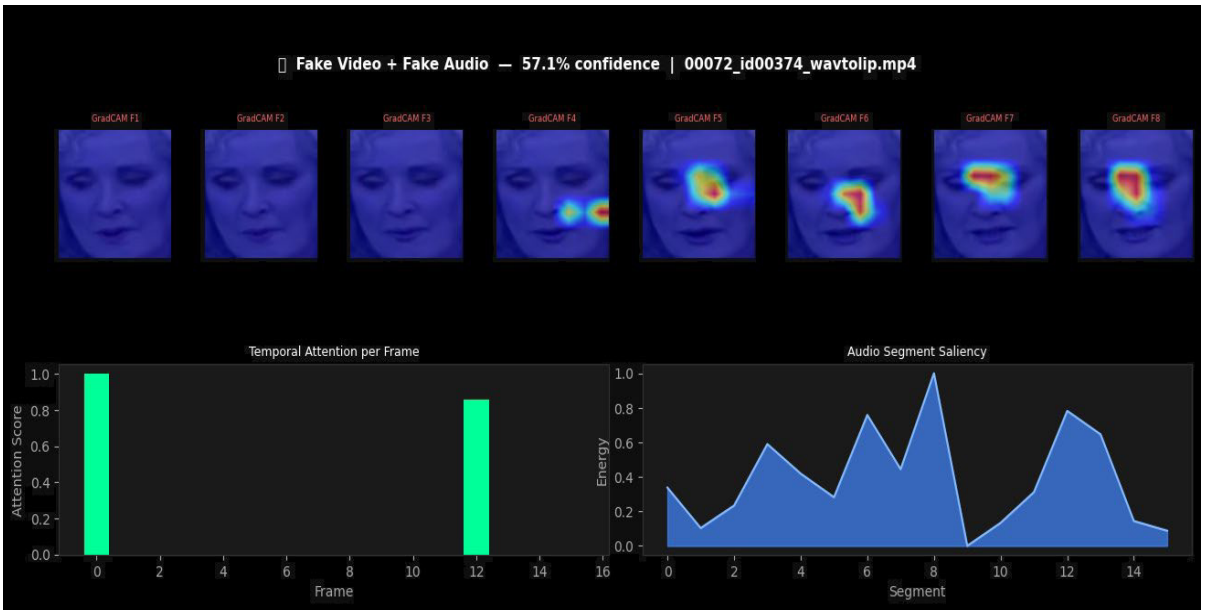

Figure 5.2 Model Prediction and Explainability Outputs

5.2 TYPES OF TESTING

To ensure comprehensive validation, multiple levels of testing were performed, including Unit Testing, Integration Testing, and Functional Testing.

5.2.1 UNIT TESTING

To validate the functionality of the various parts, unit testing was conducted for the modules of the system. Each module was individually tested to check whether there are any issues with its functionality. For the preprocessing module, unit tests were run to make sure the facial frames and audio data extraction is carried out correctly. Testing was done by checking the dimensions of the frames sampled and the mel-spectrograms generated by using videos as input for the test samples. For the visual feature extraction module, unit testing was done by passing sample frames through the ResNet encoder, which will produce the corresponding feature vectors. Similarly, the audio encoder underwent unit tests, whereby spectrograms of the input videos were used to test the feature vectors extracted.

Tests were run for the classification module as well. Tests involved the use of sample vectors passed into the model to see how it responded and output probabilities for all four categories. Particular care was taken when it came to edge cases, whereby audio data was not provided for testing; zero-valued tensor was passed instead. Finally, explainability mechanisms, like Grad-CAM and temporal attention, had unit tests as well.

5.2.2 INTEGRATION TESTING

Testing of integration was carried out in order to check for integration among various parts within the model to be built. The aim was to ensure smooth transition of data from preprocessing to the generation of outputs. For checking integration between the pre-processing and feature extraction modules, complete video input samples were provided in order to test whether both visual and audio pipelines were getting data correctly. To check integration between the visual encoder and the audio encoders, and the cross- modal transformer, synchronization score calculation was checked. The integration of the classification module and the explainability pipeline was also tested for successful generation of meaningful output after providing appropriate inputs to the classification module. Moreover, integration between the input of the entire pipeline and its output of classification and forensic reports was also checked.

5.2.3 FUNCTIONAL TESTING

Functional tests were done to check how the model operates in real-world situations. These tests included evaluation of the system with a variety of videos like real videos, deepfake videos manipulated using video alone, deepfake videos manipulated using audio alone, and deepfakes entirely manipulated. The system was also subjected to multiple datasets to gauge its generalization ability. Classification results were compared to the ground truth label for accuracy. Subtle and partial manipulations were used in certain special scenarios to see how sensitive the system is. The process of analyzing the explainability module involved analyzing the generated heat map, temporal attention scores, and audio saliency maps. Forensic analysis was conducted on the forensic report generated by the system. The system successfully passed the functional test because it generated accurate predictions and explanations.

5.3 TEST RESULTS

From the analysis, the results of the experiments conducted reveal the efficiency and effectiveness of the proposed multimodal deepfake detection model. With regard to the classification performance, the model was very efficient because it achieved accuracies of about 90.4%, 86.3%, and 83.1% on FakeAVCeleb, LAV-DF, and FaceForensics++ datasets respectively. The outcomes indicate that the proposed model can generalize on other data. Through the ROC-AUC analyses, the model was able to score 0.980, 0.949, and 0.852 on FakeAVCeleb, LAV-DF, and FaceForensics++ datasets respectively. The ROC-AUC scores give an insight into the model's ability to perform classification of the data.

In addition, the equal error rates were very low for the proposed model whereby the EER was found to be 0.9%, 0.0%, and 17.9% on FakeAVCeleb, LAV-DF, and FaceForensics++ datasets respectively. From the equal error rates, it was evident that the model was able to achieve very low errors in predicting whether a sample was fake. The per-class F1 scores revealed similar performance by the model across the four classes. The results show that the proposed model had good ability to identify audio and video manipulation to detect inconsistency between the two modalities.

```
==================================================================
  FakeAVCeleb  —  Accuracy: 90.38%
==================================================================
                      precision    recall  f1-score   support

           Real/Real     0.9333    0.9245    0.9289       106
FakeVideo/RealAudio      0.8815    0.9040    0.8926      1604
RealVideo/FakeAudio      0.8687    0.9247    0.8958        93
           Fake/Fake     0.9229    0.9016    0.9121      1992

            accuracy                         0.9038      3795
           macro avg     0.9016    0.9137    0.9074      3795
        weighted avg     0.9044    0.9038    0.9039      3795


==================================================================
  LAV-DF  —  Accuracy: 86.25%
==================================================================
                      precision    recall  f1-score   support

           Real/Real     1.0000    0.9875    0.9937        80
FakeVideo/RealAudio      0.9508    0.7250    0.8227        80
RealVideo/FakeAudio      0.8293    0.8500    0.8395        80
           Fake/Fake     0.7245    0.8875    0.7978        80

            accuracy                         0.8625       320
           macro avg     0.8761    0.8625    0.8634       320
        weighted avg     0.8761    0.8625    0.8634       320


==================================================================
  FF++  —  Accuracy: 83.12%
==================================================================
                      precision    recall  f1-score   support

           Real/Real     0.6591    0.7250    0.6905        40
FakeVideo/RealAudio      0.9123    0.8667    0.8889       120
RealVideo/FakeAudio      0.0000    0.0000    0.0000         0
           Fake/Fake     0.0000    0.0000    0.0000         0

            accuracy                         0.8313       160
           macro avg     0.3928    0.3979    0.3948       160
        weighted avg     0.8490    0.8313    0.8393       160
```

Figure 5.3 Model Prediction Output and Class Probabilities

CHAPTER 6

RESULTS AND DISCUSSION

In this chapter, we describe the analysis of the proposed multimodal deepfake detection framework, which uses both audio and visual modalities in a deep neural network architecture. For evaluating the generalization capability of our method, the experiments were conducted using several datasets that include FakeAVCeleb, LAV-DF, and FaceForensics++.

The metrics used for analyzing the performance of the model include Accuracy, ROC-AUC, EER, and F1-score. These were evaluated based on the results obtained from training analysis as well as testing on independent test datasets. Moreover, the effectiveness of the proposed multimodal technique for detecting complex deepfakes was examined by comparing it with existing single modality techniques.

6.1 SYSTEM PERFORMANCE AND EFFICIENCY

The multimodal learning framework presented was successful in generating discriminative audio-visual data throughout all tested data sets. While learning took place, there was continuous reduction of loss alongside consistent increase of accuracy, signifying stability during the learning process.

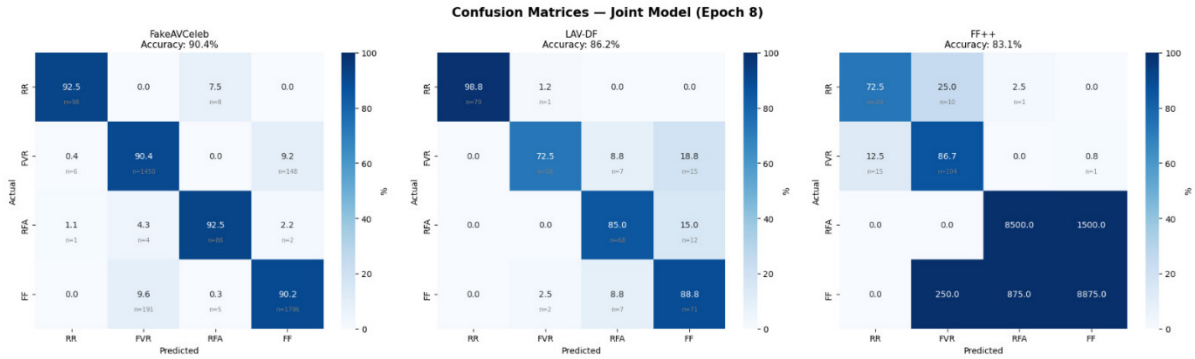

Figure 6.1 Confusion Matrix

From the classification performance, it is evident that the model performed quite well. Specifically, the model gave an accuracy score of 90.4% for FakeAVCeleb, 86.3% for LAV-DF, and 83.1% for FaceForensics++. This shows that the model generalized successfully across all three datasets, despite them having different qualities. The relatively poor performance for the latter is attributed to the

presence of heavy compression artefacts and absence of audio information in the dataset.

Figure 6.2 depicts the ROC-AUC score, the findings from our experiment confirm the model's ability to perform effectively. Particularly, a macro-AUC value of 0.980 for FakeAVCeleb and 0.949 for LAV-DF implies excellent discriminatory capabilities of the model. For FaceForensics++, the model managed a macro-AUC score of 0.852.

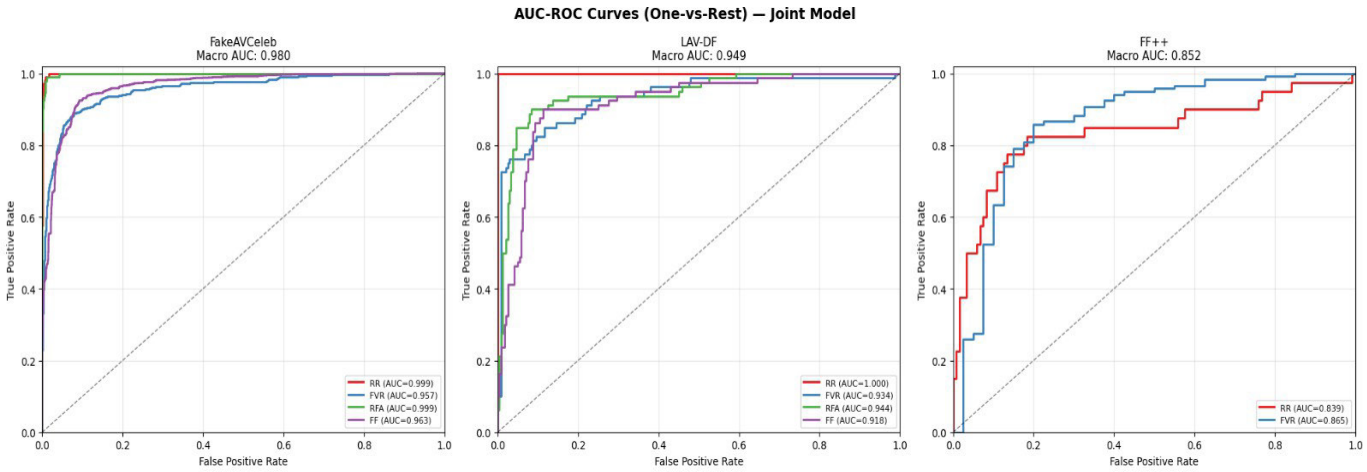

Figure 6.2 ROC–AUC Curves Across Datasets

Also, the Equal Error Rate (EER) test further confirms these findings, with an EER rate of 0.9% for FakeAVCeleb and 0.0% for LAV-DF. The results indicate very high accuracy in classifying the datasets under optimal thresholding. On the other hand, for FaceForensics++, the EER rate was 17.9%.

In conclusion, the results indicate that the integration of audiovisual data in detecting deepfake videos significantly improves performance by identifying inconsistencies that would be difficult to detect using a single-modality approach.

6.2 COMPARISON: EXISTING vs PROPOSED SYSTEM

For the assessment of the efficiency of the newly proposed hybrid model, a comparitive analysis is conducted between the current Suricata only model and the hybrid model with deep learning autoencoder included. The following table outlines the main differences based on test results:

Table 6.1: Comparison between existing and proposed system

Parameter	Existing Systems (Single Modality)	Proposed System (Multimodal)
Detection Approach	Visual or Audio Only	Audio + Visual Fusion
Generalization	Poor across datasets	85.6%
Explainability	Limited (Black-box)	High (Grad-CAM + XAI + LLM)
Manipulation Detection	Limited to specific type	Detects audio, video, and combined manipulations
Accuracy	~70–80% (dataset- dependent)	Up to 90.4%
Robustness	Low for unseen data	High for unseen manipulations

As seen from the above table 6.1, traditional deepfake detection approaches based on one modality perform poorly at detecting complex manipulations. They also show poor generalization capabilities across different datasets. However, the multimodal approach that is suggested performs well in combining the audio-visual modality and detects any discrepancy.

Moreover, adding explainability to this model makes the proposed approach transparent enough for practical applications such as digital forensics or media validation. While there is a slight increase in computational complexity compared to the traditional approach, the advantage in terms of detection capability and interpretability make up for it. In summary, this study proves the feasibility of the proposed approach.

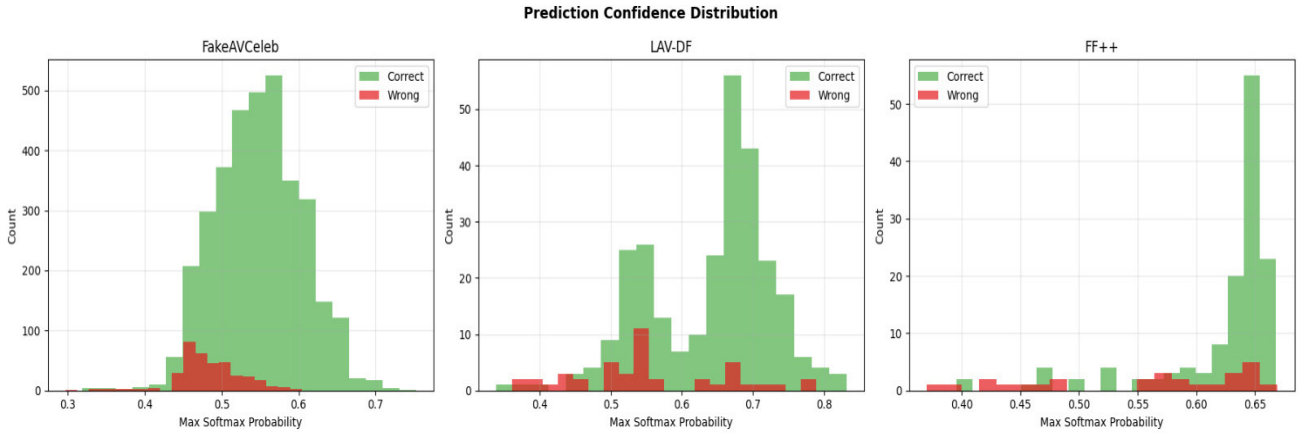

Figure 6.2 Prediction Confidence Distribution

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

7.1 CONCLUSION

The principal purpose of this work is developing an innovative detector of deepfakes based on a multi-modal model using both visual and audio modalities to improve detection performance and generalizability of the proposed approach. In particular, a new deep learning method using convolutional neural networks to process visual data and transformer to analyze temporal data is proposed. At the same time, a cross-modal attention technique is used to find possible discrepancies between visual and audio modalities. Using a number of different datasets like FakeAVCeleb, LAV-DF, and FaceForensics++ the model achieves great performance both in detecting individual and combined modifications. The use of Explainable AI techniques in the developed model gives additional benefits because it allows for understanding the process of forensic investigation. Thus, this research has addressed many issues associated with modern detectors of deepfakes.

7.2 FUTURE ENHANCEMENTS

While the proposed architecture has achieved outstanding results, there still exists much scope for improving the robustness and scalability of the system. Further research could involve identifying highly compressed videos through a compression-aware strategy during training or adopting a different backbone for processing the visual stream. The performance of the audio branch can be optimized through self-supervised learning frameworks such as Wav2Vec, which is adept at extracting audio features from deepfake videos. Model optimization can be done to allow real-time execution in various computing environments through techniques such as model quantization and knowledge distillation, thereby achieving scalability of the pipeline. The generalizability of the system can be enhanced by collecting data related to new approaches used to generate deepfake videos.

7.3 RESULTS

To evaluate the proposed multimodal deepfake detection system, several experiments were conducted using different datasets, focusing on assessing the accuracy, robustness, and generalizability of the method. On the FakeAVCeleb, LAV-DF, and FaceForensics++ datasets, the method achieved 90.4%, 86.3%, and 83.1% accuracy, respectively, yielding an average accuracy of 86.6%.

ROC-AUC analysis revealed strong discriminative power with the following values: 0.980 on the FakeAVCeleb, 0.949 on the LAV-DF, and 0.852 on the FaceForensics++. Moreover, the Equal Error Rate (EER) values supported the results, being equal to 0.9% on the FakeAVCeleb, 0.0% on the LAV-DF, and 17.9% on the FaceForensics++.

Per-class evaluation indicated balanced performance on all manipulation classes. In particular, the method achieved high F1 scores for real, audio-manipulated, video-manipulated, and fully manipulated samples. Besides, the ability of the proposed method to detect cross-modal discrepancies, that were not detected by previous single-modality models, is also notable.

Finally, the explainability module provided valuable explanations in the form of visual and textual outputs, making the method more transparent and appropriate for practical usage in forensics. Therefore, based on the experimental results, one can conclude about the significant improvement in deepfake detection performance of the proposed multimodal framework.

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APPENDIX

A. SOURCE CODE

```
def detect_face_box(frame):
    try:
        faces = RetinaFace.detect_faces(frame)
        if isinstance(faces, dict) and len(faces) > 0:
            largest = max(
                faces.values(),
                key=lambda x: (x['facial_area'][2] - x['facial_area'][0]) *
                              (x['facial_area'][3] - x['facial_area'][1])
            )
            return largest['facial_area']
    except:
        pass
    return None

def extract_frames(video_path):
    cap = cv2.VideoCapture(video_path)
    frames, face_box, frame_count = [], None, 0
    while True:
        ret, frame = cap.read()
        if not ret:
            break
        frame_count += 1
        if face_box is None and frame_count <= 30:
            face_box = detect_face_box(frame)
        if face_box is not None:
            x1, y1, x2, y2 = face_box
            h, w, _ = frame.shape
            crop = frame[max(0,y1):min(h,y2), max(0,x1):min(w,x2)]
        else:
            h, w, _ = frame.shape
            s = min(h, w) // 2
            crop = frame[h//2-s:h//2+s, w//2-s:w//2+s]
        frames.append(cv2.resize(cv2.cvtColor(crop, cv2.COLOR_BGR2RGB), (FRAME_SIZE, FRAME_SIZE)))
        if len(frames) >= NUM_FRAMES:
            break
```

```
    cap.release()
    while len(frames) < NUM_FRAMES:
        frames.append(frames[-1] if frames else np.zeros((FRAME_SIZE, FRAME_SIZE, 3), dtype=np.uint8))
    return frames

def extract_audio(video_path, tmp_path='/content/tmp_audio.wav'):
    """Extract audio. Returns (path, has_audio) tuple."""
    try:
        ffmpeg.input(video_path).output(tmp_path, ac=1, ar=SR).overwrite_output().run(quiet=True)
        if os.path.exists(tmp_path) and os.path.getsize(tmp_path) > 1000:
            return tmp_path, True
    except:
        pass
    return tmp_path, False

def process_audio(path, has_audio=True, video_path=None):
    """Process audio into mel spectrogram segments.
    If no audio, returns silence so model uses video signal only.
    """
    if not has_audio:
        print('⚠️  No audio track detected — using silence (model will rely on video only)')
        # Estimate duration from video
        if video_path:
            cap = cv2.VideoCapture(video_path)
            fps   = cap.get(cv2.CAP_PROP_FPS) or 25
            total = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
            cap.release()
            audio = np.zeros(int(SR * total / fps), dtype=np.float32)
        else:
            audio = np.zeros(SR * 5, dtype=np.float32)  # 5s silence
        sr = SR
    else:
        audio, sr = librosa.load(path, sr=SR)

    if len(audio) < NUM_AUDIO_SEG:
        audio = np.pad(audio, (0, NUM_AUDIO_SEG - len(audio)))

    seg_len = len(audio) // NUM_AUDIO_SEG
```

```
    feats   = []
    for i in range(NUM_AUDIO_SEG):
        seg = audio[i*seg_len:(i+1)*seg_len]
        mel = librosa.power_to_db(librosa.feature.melspectrogram(y=seg, sr=sr, n_mels=N_MELS))
        mel = np.pad(mel, ((0,0),(0,max(0,TARGET_MEL_T-mel.shape[1]))), mode='constant')[:,
:TARGET_MEL_T]
        feats.append(mel)
    return np.array(feats)
```

B . SCREENSHOTS

B.1 Explainability Visualization of Deepfake Detection Results

C. PUBLICATIONS DETAILS

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Multimodal Deepfake Detection with C...

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